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CONTROL DEVICE, ELECTRONIC DEVICE, AND CONTROL METHOD CAPABLE OF REDUCING NUMBER OF OUTPUT TERMINALS OF PROCESSOR EXECUTING DATA COMMUNICATION WITH INTEGRATED CIRCUIT

Abstract

A control device including a processor and a signal output circuit. The processor has an output terminal for outputting a clock signal, and executes data communication with an integrated circuit. The signal output circuit outputs a reset signal used for resetting the integrated circuit in response to an input of the clock signal having a predetermined first value of duty ratio after input of the clock signal to the integrated circuit, and stops output of the reset signal in response to switching of the duty ratio of the clock signal from the first value to a second value lower than the first value.

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Background/Summary

INCORPORATION BY REFERENCE

[0001] This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2024-033457 filed on Mar. 6, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

[0002] The present disclosure relates to a control device, an electronic device, and a control method.

[0003] Electronic devices such as printers include a processor such as a CPU. In the electronic device, data communication is executed between an integrated circuit such as an image processing circuit and the processor. For example, when data communication is executed between the processor and the integrated circuit, the processor inputs a clock signal to the integrated circuit, and inputs a reset signal to the integrated circuit after the clock signal becomes stable. In addition, the processor stops inputting the reset signal to the integrated circuit after data communication with the integrated circuit ends, and thereafter stops inputting the clock signal to the integrated circuit.

SUMMARY

[0004] A control device according to one aspect of the present disclosure includes a processor and a signal output circuit. The processor has an output terminal for outputting a clock signal, and executes data communication with an integrated circuit. The signal output circuit outputs a reset signal used for resetting the integrated circuit in response to an input of the clock signal having a predetermined first value of duty ratio after input of the clock signal to the integrated circuit, and stops output of the reset signal in response to switching of the duty ratio of the clock signal from the first value to a second value lower than the first value.

[0005] An electronic device according to another aspect of the present disclosure includes the control device and the integrated circuit.

[0006] A control method according to another aspect of the present disclosure is executed by a control device including: a processor configured to have an output terminal that outputs a clock signal, and to execute data communication with an integrated circuit; and a signal output circuit configured to output a reset signal used for resetting the integrated circuit in response to an input of the clock signal having a duty ratio of a predetermined first value after the input of the clock signal to the integrated circuit, and to stop output of the reset signal in response to switching of the duty ratio of the clock signal from the first value to a second value lower than the first value; and the control method includes an output step and a switching step. In the output step, the clock signal having a duty ratio of the first value is outputted from the output terminal when the data communication is executed. In the switching step, the duty ratio of the clock signal output from the output terminal is switched from the first value to the second value after the data communication ends.

[0007] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description with reference where appropriate to the accompanying drawings. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a cross-sectional view showing a configuration of an image forming apparatus of an embodiment according to the present disclosure.

[0009] FIG. **2** is a block diagram showing a configuration of a control portion of an embodiment according to the present disclosure.

[0010] FIG. **3** is a circuit diagram showing a configuration of an additional relay circuit of a control portion of an embodiment according to the present disclosure.

[0011] FIG. **4** is a flowchart showing an example of a communication control process executed by a CPU of the control portion of an embodiment according to the present disclosure.

DETAILED DESCRIPTION

[0012] Hereinafter, embodiments according to the present disclosure will be described with reference to the accompanying drawings. Note that the following embodiments are examples according to the present disclosure and do not limit the technical scope of the present disclosure.

[Configuration of Image Forming Apparatus **100**]

[0013] First, a configuration of an image forming apparatus **100** of an embodiment according to the present disclosure will be described with reference to FIGS. **1** and **2**.

[0014] The image forming apparatus **100** is a multifunction peripheral having multiple functions, such as a scanning function for reading an image of a document, a printing function for forming an image based on image data, a facsimile function, and a copying function. The image forming apparatus **100** is an example of an electronic device according to the present disclosure. Note that the technique according to the present disclosure may be applied to electronic devices such as scanners, printers, fax machines, copiers, personal computers, notebook computers, televisions, microwave ovens, and refrigerators.

[0015] As shown in FIG. **1**, the image forming apparatus **100** includes an auto document feeder (ADF) **1**, an image reading portion **2**, an image forming portion **3**, and a sheet feed portion **4**. In addition, the image forming apparatus **100** also includes an image processing circuit **5** and a control portion **6** shown in FIG. **2**. The control portion **6** is an example of a control device according to the present disclosure.

[0016] The ADF **1** conveys a document to be read by the scanning function. The ADF **1** includes a document setting portion, a plurality of document conveying rollers, a document holder, and a sheet discharge portion.

[0017] The image reading portion **2** achieves the scanning function. The image reading portion **2** includes a document table, a light source, a plurality of mirrors, an optical lens, and a charge coupled device (CCD).

[0018] The image forming portion **3** achieves the printing function. As shown in FIG. **1**, the image forming portion **3** includes a photoconductor drum **11**, a charging roller **12**, a laser scanning unit **13**, a developing device **14**, a toner container **15**, a transfer roller **16**, a cleaning device **17**, a fixing device **18**, and a sheet discharge tray **19**.

[0019] The photoconductor drum **11** is rotatably provided. The charging roller **12** is provided in contact with a surface of the photoconductor drum **11** and charges the surface of the photoconductor drum **11**.

[0020] The laser scanning unit **13** irradiates the surface of the photoconductor drum **11**, which has been charged by the charging roller **12**, with light based on image data. The laser scanning unit **13** forms an electrostatic latent image on the surface of the photoconductor drum **11**.

[0021] The developing device **14** develops the electrostatic latent image formed on the surface of the photoconductor drum **11** with toner. The toner container **15** supplies toner to the developing device **14**.

[0022] The transfer roller **16** transfers the electrostatic latent image (toner image) developed by the developing device **14** onto a sheet supplied by the sheet feed portion **4**. The cleaning device **17** cleans the surface of the photoconductor drum **11** after the toner image has been transferred by the transfer roller **16**.

[0023] The fixing device **18** fixes the toner image transferred onto the sheet by the transfer roller **16** to the sheet. The sheet to which the toner image has been fixed by the fixing device **18** is discharged to the sheet discharge tray **19**.

[0024] The sheet feed portion **4** supplies sheets to the image forming portion **3**. As shown in FIG. **1**, the sheet feed portion **4** includes a sheet feed cassette **21**, a pickup roller **22**, a sheet feed roller **23**, a plurality of sheet conveying rollers **24**, and a registration roller **25**.

[0025] The sheet feed cassette **21** is detachably provided in a housing of the image forming apparatus **100**, and sheets to be fed to the image forming unit **3** are placed therein. The pickup roller **22** picks up the topmost sheet among sheets placed in the sheet feed cassette **21** from the sheet feed cassette **21**.

[0026] The sheet feed roller **23** conveys the sheet picked up from the sheet feed cassette **21** by the pickup roller **22** to a sheet supply path leading to the image forming portion **3**. A plurality of sheet conveying rollers **24** are provided in the sheet supply path and convey the sheet to the image forming portion **3**.

[0027] The registration roller **25** is provided at the most downstream portion of the sheet supply path, and supplies the sheet to the image forming portion **3** at a predetermined timing.

[0028] The image processing circuit **5** executes predetermined image processing on the image data to be printed. For example, the image processing circuit **5** is an application specific integrated circuit (ASIC), a digital signal processor (DSP), or a field programmable gate array (FPGA). The image processing circuit **5** is an example of an integrated circuit according to the present disclosure. Note that the integrated circuit according to the present disclosure may be a storage device such as a flash memory. In addition, the integrated circuit according to the present disclosure may be an engine control portion that controls the image forming portion **3** and the sheet feed portion **4**.

[0029] The control portion **6** performs overall control of the image forming apparatus **100**. Note that the control portion **6** may be the engine control portion.

[0030] As shown in FIG. **2**, the control portion **6** includes a CPU **30**.

[0031] The CPU **30** is a processor that executes various types of arithmetic processing. The CPU **30** executes data communication with the image processing circuit **5** via a data transmission path (not shown). The CPU **30** is an example of a processor according to the present disclosure.

[0032] In the image forming apparatus **100**, when data communication is started between the image processing circuit **5** and the CPU **30**, a clock signal X**2** (see FIG. **2**) is input to the image processing circuit **5**, and then a reset signal X**3** (see FIG. **2**) is input to the image processing circuit **5**. In addition, after data communication between the image processing circuit **5** and the CPU **30** ends, the input of the reset signal X**3** to the image processing circuit **5** is stopped, and thereafter the input of the clock signal X**2** to the image processing circuit **5** is stopped.

[0033] In conventional electronic devices, in order to achieve data communication between the CPU **30** and the image processing circuit **5**, it is necessary to provide the CPU **30** with separate output terminals used for outputting the clock signal X**2** and for outputting the reset signal X**3**.

[0034] However, in the image forming apparatus **100** according of this embodiment according to the present disclosure, as will be described below, it is possible to reduce the number of output terminals of the CPU **30** that executes data communication with the image processing circuit **5**.

[0035] More specifically, the CPU **30** includes an output terminal **31** (see FIG. **2**) that outputs a clock signal X**1** (see FIG. **2**).

[0036] In addition, the control portion **6** also includes an additional relay circuit **40** shown in FIG. **2** and FIG. **3**.

[0037] Furthermore, the CPU **30** also includes an output processing portion **32**, a switching processing portion **33**, and a stop processing portion **34** shown in FIG. **2**. More specifically, the

CPU **30** functions as each of the above-mentioned processing portions by executing a control program stored in advance in a ROM (not shown).

[Configuration of Additional Relay Circuit **40**]

[0038] Next, a configuration of the additional relay circuit **40** will be described with reference to FIG. **2** and FIG. **3**.

[0039] As shown in FIG. **3**, the additional relay circuit **40** includes a clock signal output circuit **41** and a reset signal output circuit **42**.

[0040] In response to input of the clock signal X**1**, the clock signal output circuit **41** generates and outputs a clock signal X**2** that is identical to the clock signal X**1**.

[0041] As shown in FIG. **3**, the clock signal output circuit **41** includes resistors R**1** to R**3** and a transistor Q**1**.

[0042] The transistor Q**1** is a PNP type transistor. The emitter terminal of the transistor Q**1** is connected to a power supply Vcc (see FIG. **3**) via the resistor R**1**. In addition, the emitter terminal of the transistor Q**1** is connected to an input terminal **51** of the image processing circuit **5** (see FIG. **2**). The input terminal **51** receives the clock signal X**2**. A base terminal of the transistor Q**1** is connected to an output terminal **31** of the CPU **30** (see FIG. **2**) via the resistor R**2**. In addition, the base terminal of the transistor Q**1** is connected to ground via the resistor R**3**. A collector terminal of the transistor Q**1** is connected to ground.

[0043] In the clock signal output circuit **41**, the transistor Q**1** is in an ON state when the clock signal X**1** is not output from the output terminal **31** (see FIG. **2**) of the CPU **30**. When the transistor Q**1** is in the ON state, the input terminal **51** (see FIG. **2**) of the image processing circuit **5** is connected to ground. In other words, when the clock signal X**1** is not being output from the output terminal **31** of the CPU **30**, the clock signal output circuit **41** does not generate the clock signal X**2**.

[0044] In the clock signal output circuit **41**, the transistor Q**1** is in an OFF state when the clock signal X**1** is output from the output terminal **31** (see FIG. **2**) of the CPU **30** and the signal level of the clock signal X**1** is at a high level. When the transistor Q**1** is in the OFF state, the input terminal **51** (see FIG. **2**) of the image processing circuit **5** is connected to the power supply Vcc via the resistor R**1**. Thus, a high-level signal of the clock signal X**2** is output. In addition, in the clock signal output circuit **41**, the transistor Q**1** is in an ON state when the clock signal X**1** is output from the output terminal **31** of the CPU **30** and the signal level of the clock signal X**1** is at a low level. When the transistor Q**1** is in the ON state, the input terminal **51** of the image processing circuit **5** is connected to ground. Thus, a low-level signal of the clock signal X**2** is output.

[0045] The reset signal output circuit **42** outputs a reset signal X**3** after the clock signal X**2** is input to the image processing circuit **5** in response to the input of a clock signal X**1** having a duty ratio with a predetermined first value. In addition, the reset signal output circuit **42** stops outputting the reset signal X**3** in response to switching the duty ratio of the clock signal X**1** from the first value to a second value lower than the first value. The reset signal output circuit **42** is an example of a signal output circuit according to the present disclosure.

[0046] As shown in FIG. **3**, the reset signal output circuit **42** includes resistors R**4** to R**10**, transistors Q**2** to Q**3**, and capacitors C**1** and C**2**.

[0047] The transistor Q**2** is an NPN type transistor. A collector terminal of the transistor Q**2** is connected to the power supply Vcc via the resistor R**4**. A base terminal of the transistor Q**2** is connected to the output terminal **31** of the CPU **30** (see FIG. **2**) via the resistor R**5**. An emitter terminal of the transistor Q**2** is connected to ground via the resistor R**6**. In addition, the emitter terminal of the transistor Q**2** is connected to ground via the capacitor C**1**.

[0048] The transistor Q**3** is a PNP type transistor. An emitter terminal of the transistor Q**3** is connected to the power supply Vcc via the resistor R**7**. In addition, the emitter terminal of the transistor Q**3** is connected to the input terminal **51** of the image processing circuit **5** (see FIG. **2**). The reset signal X**3** is input to the input terminal **52**. A base terminal of the transistor Q**3** is connected to the emitter terminal of the transistor Q**2** via the resistor R**8**. In addition, the base

terminal of the transistor Q**3** is connected to ground via the resistor R**9**. A collector terminal of the transistor Q**3** is connected to ground.
[0049] In addition, the emitter terminal of the transistor Q**3** is connected to ground via the capacitor C**2**. Furthermore, the emitter terminal of the transistor Q**3** is connected to ground via the resistor R**10**.
[0050] In the reset signal output circuit **42**, the transistor Q**2** is in an OFF state when the clock signal X**1** is not output from the output terminal **31** of the CPU **30** (see FIG. **2**). That is, no voltage is applied to the base terminal of the transistor Q**3**. Therefore, when the clock signal X**1** is not output from the output terminal **31** of the CPU **30**, the transistor Q**3** is in an ON state. When the transistor Q**3** is in the ON state, the input terminal **52** (see FIG. **2**) of the image processing circuit **5** is connected to ground. In other words, when the clock signal X**1** is not being output from the output terminal **31** of the CPU **30**, the reset signal output circuit **42** does not generate the reset signal X**3**.
[0051] In the reset signal output circuit **42**, the transistor Q**2** is in an ON state when the clock signal X**1** is output from the output terminal **31** (see FIG. **2**) of the CPU **30** and the signal level of the clock signal X**1** is at a high level. When the transistor Q**2** is in the ON state, the capacitor C**1** is connected to the power supply Vcc via the resistor R**4**. Thus, the capacitor C**1** is charged, causing the voltage applied to the base terminal of the transistor Q**3** to gradually increase, switching the transistor Q**3** from an ON state to an OFF state. When the transistor Q**3** is in the OFF state, the capacitor C**2** is connected to the power supply Vcc via the resistor R**7**. Thus, the capacitor C**2** is charged, and the voltage applied to the input terminal **52** (see FIG. **2**) of the image processing circuit **5** gradually increases.
[0052] In the reset signal output circuit **42**, the transistor Q**2** is in an OFF state when the clock signal X**1** is output from the output terminal **31** (see FIG. **2**) of the CPU **30** and the signal level of the clock signal X**1** is at a low level. When the transistor Q**2** is OFF, the capacitor C**1** discharges. Thus, the transistor Q**3** remains in the OFF state for a while.
[0053] Here, the higher the duty ratio of the clock signal X**1**, the longer the charging time of the capacitor C**1** becomes and the shorter the discharging time of the capacitor C**1** becomes. In other words, when the duty ratio of the clock signal X**1** is sufficiently high, it is possible to maintain the transistor Q**3** in the OFF state regardless of the switching of the signal level of the clock signal X**1**. Therefore, it is possible to increase the voltage applied to the input terminal **52** (see FIG. **2**) of the image processing circuit **5** up to the signal level of the reset signal X**3**.
[0054] More specifically, in the reset signal output circuit **42**, when the duty ratio of the clock signal X**1** is equal to or greater than a third value, it is possible to maintain the transistor Q**3** in the OFF state regardless of switching of the signal level of the clock signal X**1**. In addition, the first value is a value set within a range equal to or greater than the third value.
[0055] On the other hand, as the duty ratio of the clock signal X**1** decreases, the charging time of the capacitor C**1** decreases and the discharging time of the capacitor C**1** increases. In other words, when the duty ratio of the clock signal X**1** is sufficiently low, the transistor Q**3** switches from the OFF state to the ON state during the period in which the signal level of the clock signal X**1** is at the low level, and the capacitor C**2** discharges. When the discharged amount of the capacitor C**2** exceeds the charged amount of the capacitor C**2**, the voltage applied to the input terminal **52** (see FIG. **2**) of the image processing circuit **5** gradually decreases. In this case, the output of the reset signal X**3** stops.
[0056] More specifically, in the reset signal output circuit **42**, when the duty ratio of the clock signal X**1** is equal to or lower than a fourth value that is lower than the third value, the discharge amount of the capacitor C**2** exceeds the charge amount of the capacitor C**2**. In addition, the second value is a value set within a range equal to or less than the fourth value.
[Functions of Each Processing Portion Included in the CPU **30**]
[0057] Next, the function of each processing portion included in the CPU **30** will be described with

reference to FIG. **2**.

[0058] When data communication is performed between the CPU **30** and the image processing circuit **5**, the output processing portion **32** causes the output terminal **31** to output the clock signal X**1** having a duty ratio of the first value.

[0059] After data communication between the CPU **30** and the image processing circuit **5** ends, the switching processing portion **33** switches the duty ratio of the clock signal X**1** output from the output terminal **31** from the first value to the second value.

[0060] The stop processing portion **34** stops the output of the clock signal X**1** from the output terminal **31** after the data communication between the CPU **30** and the image processing circuit **5** is completed and the output of the reset signal X**3** from the reset signal output circuit **42** is stopped.

[Communication Control Process]

[0061] Hereinafter, an example of a procedure of the communication control process executed by the CPU **30** in the image forming apparatus **100** and the control method according to the present disclosure will be described with reference to FIG. **4**. Here, steps S**11**, S**12**, . . . represent numbers of processing procedures (steps) executed by the CPU **30**. Note that the communication control process is executed when data communication is performed between the CPU **30** and the image processing circuit **5**.

<Step S**11**>

[0062] First, in step S**11**, the CPU **30** causes the output terminal **31** to output the clock signal X**1** having a duty ratio of the first value. The process of step S**11** is an example of an output step according to the present disclosure, and is executed by the output processing portion **32** of the CPU **30**.

[0063] By executing the process of step S**11**, the clock signal X**1** having the duty ratio of the first value is input to the additional relay circuit **40** (see FIG. **2**). Thus, the clock signal output circuit **41** outputs the clock signal X**2**. In addition, after the clock signal X**2** is input to the image processing circuit **5**, the reset signal output circuit **42** outputs the reset signal X**3**.

<Step S**12**>

[0064] In step S**12**, the CPU **30** determines whether or not the reset signal X**3** has been output from the reset signal output circuit **42**.

[0065] For example, the CPU **30** determines that the reset signal X**3** has been output from the reset signal output circuit **42** when a predetermined first time has elapsed since the execution of the process of step S**11**.

[0066] Here, when the CPU **30** determines that the reset signal X**3** has been output from the reset signal output circuit **42** (Yes in S**12**), the CPU **30** moves the process to step S**13**. In addition, when the reset signal X**3** is not being output from the reset signal output circuit **42** (No in S**12**), the CPU **30** waits for the reset signal X**3** to be output from the reset signal output circuit **42** in step S**12**.

<Step S**13**>

[0067] In step S**13**, the CPU **30** starts data communication with the image processing circuit **5**.

<Step S**14**>

[0068] In step S**14**, the CPU **30** determines whether or not data communication with the image processing circuit **5** has ended.

[0069] Here, when the CPU **30** determines that the data communication with the image processing circuit **5** has ended (Yes in S**14**), the CPU **30** moves the process to step S**15**. In addition, when the data communication with the image processing circuit **5** has not ended (No in S**14**), the CPU **30** waits for the data communication with the image processing circuit **5** to end in step S**14**.

<Step S**15**>

[0070] In step S**15**, the CPU **30** switches the duty ratio of the clock signal X**1** output from the output terminal **31** from the first value to the second value. The process of step S**15** is an example of a switching step according to the present disclosure, and is executed by the switching processing portion **33** of the CPU **30**.

[0071] By executing the process of step S**15**, the clock signal X**1** having the duty ratio of the second value is input to the additional relay circuit **40** (see FIG. **2**). Thus, the clock signal output circuit **41** outputs the clock signal X**2**. In addition, the output of the reset signal X**3** from the reset signal output circuit **42** stops.
<Step S**16**>
[0072] In step S**16**, the CPU **30** determines whether or not the output of the reset signal X**3** from the reset signal output circuit **42** has stopped.
[0073] For example, when a predetermined second time has elapsed since the execution of the process of step S**15**, the CPU **30** determines that the output of the reset signal X**3** from the reset signal output circuit **42** has stopped.
[0074] Here, when the CPU **30** determines that the output of the reset signal X**3** from the reset signal output circuit **42** has stopped (Yes in S**16**), the CPU **30** moves the process to step S**17**. In addition, when the output of the reset signal X**3** from the reset signal output circuit **42** has not stopped (No in S**16**), the CPU **30** waits for the output of the reset signal X**3** from the reset signal output circuit **42** to stop in step S**16**.
<Step S**17**>
[0075] In step S**17**, the CPU **30** stops the output of the clock signal X**1** from the output terminal **31**. The process of step S**17** is executed by the stop processing portion **34** of the CPU **30**.
[0076] Thus, it is possible to reduce power consumption and suppress the occurrence of electromagnetic interference due to the output of the clock signal X**1**, as compared to a configuration in which the CPU **30** always outputs the clock signal X**1**.
[0077] In this way, the control portion **6** includes a reset signal output circuit **42** that outputs a reset signal X**3** in response to input of the clock signal X**1** having a duty ratio of the first value after the clock signal X**2** is input to the image processing circuit **5**, and stops output of the reset signal X**3** in response to switching of the duty ratio of the clock signal X**1** to the second value. Thus, it possible to output the reset signal X**3** without providing the CPU **30** with an output terminal for the reset signal X**3**. Therefore, the number of output terminals of the CPU **30** that executes data communication with the image processing circuit **5** can be reduced.
[Supplementary Notes of the Invention]
[0078] An outline of the invention extracted from the above-described embodiments will be added below. Note that the configurations and processing functions described in the following supplementary notes can be selected and combined as desired.
<Supplementary Note 1>
[0079] A control device, including: [0080] a processor configured to have an output terminal for outputting a clock signal and to execute data communication with an integrated circuit; and [0081] a signal output circuit configured to output a reset signal used for resetting the integrated circuit in response to an input of the clock signal having a predetermined first value of duty ratio after input of the clock signal to the integrated circuit, and to stop output of the reset signal in response to switching of the duty ratio of the clock signal from the first value to a second value lower than the first value.
<Supplementary Note 2>
[0082] The control device according to Supplementary Note 1, wherein [0083] the processor outputs the clock signal having a duty ratio of the first value from the output terminal when the data communication is executed, and switches the duty ratio of the clock signal output from the output terminal from the first value to the second value after the data communication ends.
<Supplementary Note 3>
[0084] The control device according to Supplementary Note 2, wherein [0085] the processor stops output of the clock signal from the output terminal after the data communication ends and after output of the reset signal from the signal output circuit stops.
<Supplementary Note 4>

[0086] An electronic device including: [0087] the control device according to any one of Supplementary Notes 1 to 3; and [0088] the integrated circuit.
<Supplementary Note 5>
[0089] A control method executed by a control device including: a processor configured to have an output terminal that outputs a clock signal, and to execute data communication with an integrated circuit; and a signal output circuit configured to output a reset signal used for resetting the integrated circuit in response to an input of the clock signal having a duty ratio of a predetermined first value after the input of the clock signal to the integrated circuit, and to stop output of the reset signal in response to switching of the duty ratio of the clock signal from the first value to a second value lower than the first value; [0090] the control method including: [0091] an output step of outputting the clock signal having a duty ratio of the first value from the output terminal when the data communication is executed; and [0092] a switching step of switching the duty ratio of the clock signal output from the output terminal from the first value to the second value after the data communication ends.
[0093] It is to be understood that the embodiments herein are illustrative and not restrictive, since the scope of the disclosure is defined by the appended claims rather than by the description preceding them, and all changes that fall within metes and bounds of the claims, or equivalence of such metes and bounds thereof are therefore intended to be embraced by the claims.

Claims

1. A control device, comprising: a processor configured to have an output terminal for outputting a clock signal and to execute data communication with an integrated circuit; and a signal output circuit configured to output a reset signal used for resetting the integrated circuit in response to an input of the clock signal having a predetermined first value of duty ratio after input of the clock signal to the integrated circuit, and to stop output of the reset signal in response to switching of the duty ratio of the clock signal from the first value to a second value lower than the first value.
2. The control device according to claim 1, wherein the processor outputs the clock signal having a duty ratio of the first value from the output terminal when the data communication is executed, and switches the duty ratio of the clock signal output from the output terminal from the first value to the second value after the data communication ends.
3. The control device according to claim 2, wherein the processor stops output of the clock signal from the output terminal after the data communication ends and after output of the reset signal from the signal output circuit stops.
4. An electronic device comprising: the control device according to claim 1; and the integrated circuit.
5. A control method executed by a control device comprising: a processor configured to have an output terminal that outputs a clock signal, and to execute data communication with an integrated circuit; and a signal output circuit configured to output a reset signal used for resetting the integrated circuit in response to an input of the clock signal having a duty ratio of a predetermined first value after the input of the clock signal to the integrated circuit, and to stop output of the reset signal in response to switching of the duty ratio of the clock signal from the first value to a second value lower than the first value; the control method comprising: an output step of outputting the clock signal having a duty ratio of the first value from the output terminal when the data communication is executed; and a switching step of switching the duty ratio of the clock signal output from the output terminal from the first value to the second value after the data communication ends.